Capacity-Radius Proximity for Smooth-Domain Reed–Solomon Codes

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Abstract

We study list decoding and mutual correlated agreement for constant-rate Reed–Solomon codes over smooth evaluation domains. The primary received-word space is the two-fold interleaving $\mathcal{C}^{\equiv 2} \subseteq (\mathbb{F}^2)^n$. We isolate three layers that are often conflated: normalized distance of a received word from the interleaved code, worst-case list size at that radius, and the fraction of affine challenges for which a scalar projection appears close while its underlying pair does not. A field-valued two-coordinate plane is exactly an arbitrary word in $(\mathbb{F}^2)^n$, and it is equivalently the base–direction pair of an affine line of scalar received words. This gives a lossless interface between interleaved list decoding and mutual correlated agreement.

At count radius $n-k$, a direct agreement-set injection and a matching received-word construction prove

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}$$

for every positive interleaving width. Thus $\binom{n}{k} \le \varepsilon|\mathbb{F}|$ is necessary and sufficient for the requested fractional list target at the capacity endpoint. A next-shell construction then gives a complete three-regime phase split. We instantiate the safe regime with an explicit rate-1/4 smooth family over $\mathbb{F}_{3^n}$. For pair mutual correlated agreement, an injective compressed-certificate map gives the radius-independent bound

$$|\operatorname{Bad}_{\mathrm{MCA}}| \le \binom{n}{k+1}.$$

The same explicit family satisfies both $\binom{n}{k} \le 2^{-128}|\mathbb{F}|$ and $\binom{n}{k+1} \le 2^{-128}|\mathbb{F}|$. Thus its list target holds at the capacity endpoint and its pair-MCA error is at most 2^{-128} at every normalized radius. The optimal list-decoding radius in the intermediate field-size regime remains open.

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1 Introduction

The Grand List Decoding and Grand Mutual Correlated Agreement challenges ask for quantitative proximity guarantees for constant-rate Reed–Solomon codes over smooth evaluation domains [ABF26]. We work with the two-fold interleaving $\mathcal{C}^{\equiv 2} \subseteq (\mathbb{F}^2)^n$. Every received pair $p = (p_0, p_1)$ determines the scalar affine line

$$p \in (\mathbb{F}^2)^n \quad \longmapsto \quad y_\gamma = p_0 + \gamma p_1 \in \mathbb{F}^n.$$

List decoding controls the ambiguity of the full pair, while mutual correlated agreement controls the fraction of challenges for which y_γ conceals pair-farness.

At the capacity-grid radius $(n-k)/n$, we prove the exact universal list size

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}$$

for every positive interleaving width. A matching next-shell construction gives the resulting three-regime field-size phase split. For pair MCA, an injective compressed-certificate map proves the radius-independent bound

$$|\,\mathrm{Bad}_{\mathrm{MCA}}\,| \le \binom{n}{k+1}.$$

An explicit smooth rate-1/4 family over $\mathbb{F}_{3^n}$ meets both requested 2^{-128} targets. The endpoint constructions, MCA certificate bound, numerical inequalities, and generic source-to-pair calibration implication are machine-checked in Lean.

The coding theorems quantify over the complete ambient received-word and affine-line spaces. Their 2^{-128} instantiation uses the displayed exponential-size field and is not a claim about every supplied smooth code. For an application whose observations lie upstream of the pair carrier, its source metric may be defined by pullback along the extraction map. Applications retaining a different source metric must provide one explicit comparison to the pair-code distance.

2 Problem Setup

2.1 Reed–Solomon parameters

Let $\mathbb{F}$ be a finite field and let $\mathcal{L} = (a_i)_{i \in I}$ be an injectively indexed evaluation domain, where $|I| = n$. For $1 \le k \le n$, define

$$\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k] = \{(f(a_i))_{i \in I} : f \in \mathbb{F}[X],\ \deg f < k\}.$$

The rate is $\rho = k/n$. The prize-facing regime takes ρ from a fixed constant set, such as

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\right\},$$

and requires $\mathcal{L}$ to have an appropriate smooth structure.

2.2 Relative Hamming distance

For a finite alphabet A and $u, v \in A^I$, let

$$d_H(u, v) = |\{i \in I : u_i \neq v_i\}|, \qquad \Delta(u, v) = \frac{d_H(u, v)}{n}.$$

For a code $\mathcal{D} \subseteq A^I$, define

$$\Delta(w, \mathcal{D}) = \min_{c \in \mathcal{D}} \Delta(w, c).$$

All radii in the main theorem statements are relative radii in $[0, 1]$. Natural-number radii remain useful for finite rounding, but they are not the paper-facing metric.

2.3 Interleaving

For a positive integer m, define

$$\mathcal{C}^{\equiv m} = \left\{p \in (\mathbb{F}^m)^I : \text{for every } j \in \{1, \dots, m\},\ (p_{i,j})_{i \in I} \in \mathcal{C}\right\}.$$

Hamming distance on $(\mathbb{F}^m)^I$ counts a coordinate i as erroneous when the complete vector symbol differs. It does not sum componentwise scalar distances.

2.4 List-decoding balls

For $w \in (\mathbb{F}^m)^I$ and $\delta \in [0,1]$, define

$$\Lambda(\mathcal{C}^{\equiv m}, w, \delta) = \{c \in \mathcal{C}^{\equiv m} : \Delta(w,c) \le \delta\}.$$

The worst-case list size is

$$L_m(\mathcal{C}, \delta) = \max_{w \in (\mathbb{F}^m)^I} |\Lambda(\mathcal{C}^{\equiv m}, w, \delta)|.$$

The maximum over received words is essential. A bound on a structured subset of received words is an application result, not the universal list-decoding statement.

3 Planes, Pair Words, And Affine Lines

3.1 The full ambient pair space

Define a field-valued two-coordinate plane to be a function

$$p : I \longrightarrow (\{0,1\} \longrightarrow \mathbb{F}).$$

By reassociating function spaces, the plane carrier is exactly

$$I \to (\{0,1\} \to \mathbb{F}) \cong (\mathbb{F}^2)^I.$$

Thus the word "plane" introduces no restriction on the received-word space.

Proposition 3.1 (Plane–pair equivalence)**.** *The map sending a plane p to the vector word*

$$i \longmapsto (p(i,0), p(i,1))$$

is a bijection from field-valued planes to $(\mathbb{F}^2)^I$. It preserves and reflects Hamming distance.

Proof. The inverse sends a pair word w to the function

$$(i,j) \longmapsto w_i(j).$$

Both composites are pointwise identities. At coordinate i, the two planes are equal exactly when their complete pair symbols are equal, so the Hamming disagreement sets coincide. □

3.2 Pair words as affine lines

For a plane p, define its base and direction by

$$p_0(i) = p(i, 0), \qquad p_1(i) = p(i, 1).$$

The associated affine line of scalar received words is

$$\ell_p(\gamma) = p_0 + \gamma p_1, \qquad \gamma \in \mathbb{F}.$$

Proposition 3.2 (Plane–line equivalence)**.** *The map $p \mapsto (p_0, p_1)$ is a bijection between field-valued planes and affine-line presentations consisting of an arbitrary base word and arbitrary direction word in $\mathbb{F}^I$.*

Proof. Given an arbitrary pair (f_0, f_1), define

$$p(i, 0) = f_0(i), \qquad p(i, 1) = f_1(i).$$

This construction is inverse to taking base and direction. □

3.3 Challenge evaluation is lossy

For fixed γ, the map

$$(p_0, p_1) \longmapsto p_0 + \gamma p_1$$

is generally many-to-one. Consequently, pair-farness cannot be inferred deterministically from one fixed challenge without an additional good-challenge hypothesis. MCA supplies a probabilistic statement over a challenge sampled after the pair has been fixed.

4 The Capacity-Radius Regime

4.1 The finite Hamming grid

Let $1 \le k \le n$. The capacity endpoint is already an attainable normalized Hamming radius:

$$r_{\mathrm{cap}} = n - k, \qquad \delta_{\mathrm{cap}} = \frac{n - k}{n} = 1 - \rho.$$

There is therefore no rounding loss at the endpoint. For example, the rate-1/4, length-4096 member has

$$k = 1024, \qquad r_{\mathrm{cap}} = 3072, \qquad \delta_{\mathrm{cap}} = \frac{3}{4}.$$

The previously useful strict checkpoint 3071/4096 is subsumed by the endpoint theorem below.

There is, however, a distinct granularity issue in the phrase "largest relative radius." A closed Hamming ball changes only at

$$\mathcal{H}_n = \left\{0, \frac{1}{n}, \dots, \frac{n}{n}\right\}.$$

Consequently, a safe-radius set defined over all real $\delta \in [0,1]$ may be half-open and have no largest element. We report both the largest radius in $\mathcal{H}_n$ and the supremum over real radii.

4.2 An explicit smooth large-field family

For $u \geq 0$, set

$$k_u = 64 \cdot 2^u = 2^{u+6}, \qquad n_u = 4k_u = 256 \cdot 2^u = 2^{u+8}, \qquad q_u = 3^{n_u}.$$

Let $\mathbb{F}_u = \mathbb{F}_{q_u}$. Since the multiplicative group of a finite field is cyclic of order $q_u - 1$, it suffices to check $n_u \mid (q_u - 1)$. Euler's theorem gives this directly:

$$\varphi(n_u) = \frac{n_u}{2}, \qquad \gcd(3, n_u) = 1, \qquad 3^{n_u/2} \equiv 1 \pmod{n_u}.$$

Squaring yields $3^{n_u} \equiv 1 \pmod{n_u}$. Thus $\mathbb{F}_u^\times$ contains a subgroup H_u of order n_u. We take $\mathcal{L}_u = H_u$ and

$$\mathcal{C}_u = \mathrm{RS}[\mathbb{F}_u, H_u, k_u].$$

The evaluation length and degree bound are powers of two, the domain is a multiplicative subgroup, and the rate is exactly $1/4$. This is the standard smooth-domain condition for FFT-based Reed–Solomon codes [ACFY24], and it is the explicit family used in the quantitative statement.

4.3 The field-size inequality

Fix $\varepsilon^* = 2^{-128}$. For the family above,

$$\binom{n_u}{k_u} \leq 2^{n_u}.$$

Moreover n_u is even and $n_u/2 \geq 128$, so

$$\begin{aligned} q_u = 3^{n_u} &= (3^2)^{n_u/2} = 9^{n_u/2} \\ &\geq 8^{n_u/2} = 2^{3n_u/2} \geq 2^{n_u+128}. \end{aligned}$$

It follows that

$$\binom{n_u}{k_u} \le 2^{-128} q_u = \varepsilon^* |\mathbb{F}_u|.$$

The pair-MCA theorem uses the comparison

$$2^{128} \binom{n_u}{k_u + 1} \le q_u.$$

It follows from the same calculation, because $\binom{n_u}{k_u+1} \le 2^{n_u}$. Compared with the ordered certificate count, this removes a factor $k_u + 1$:

$$\binom{n_u}{k_u}(n_u - k_u) = (k_u + 1)\binom{n_u}{k_u + 1}.$$

The field is exponential in the block length. This is a valid explicit sufficiently-large-field regime, not a claim about practically sized fields or about every smooth code supplied to the Grand Challenge.

5 The Capacity-Grid List Theorem

The following argument applies to every injectively indexed Reed–Solomon evaluation domain; smoothness enters only when we instantiate the explicit family from Section 4.2.

Theorem 5.1 (Universal capacity-grid equality)**.** *Let $\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k]$ have length n, where $1 \le k \le n$, and let $m \ge 1$. Then*

$$L_m\left(\mathcal{C}, \frac{n-k}{n}\right) = \binom{n}{k}.$$

In particular, the worst-case endpoint list size is independent of m.

Proof. We first prove the upper bound. Fix $w \in (\mathbb{F}^m)^n$. Every codeword in the ball agrees with w on at least k complete vector-symbol coordinates. Assign to each such codeword one k-subset of its agreement coordinates. This assignment is injective: if two interleaved codewords are assigned the same set S, then they agree with one another on S. In each component, two degree-$< k$ polynomials agree at the k distinct evaluation points indexed by S, and hence are equal. The two interleaved codewords are therefore equal. There are exactly $\binom{n}{k}$ possible sets S.

For the reverse inequality, choose one of the m components and define the received word

$$w_i^\star = (a_i^k, 0, \ldots, 0).$$

For each k-subset $S \subseteq I$, put

$$V_S(X) = \prod_{i \in S}(X - a_i), \qquad f_S(X) = X^k - V_S(X).$$

The polynomial V_S is monic of degree k, so its leading term cancels that of X^k and $\deg f_S < k$. Use f_S in the distinguished component and zero polynomials in all other components to obtain an interleaved codeword c_S.

At evaluation coordinate i,

$$c_{S,i} = w_i^\star \quad \Longleftrightarrow \quad V_S(a_i) = 0 \quad \Longleftrightarrow \quad i \in S,$$

where the final equivalence uses injectivity of the evaluation points. Consequently c_S has distance exactly $n-k$ from $w^\star$, and its agreement set recovers S. The $\binom{n}{k}$ choices of S therefore give distinct codewords in the endpoint ball. This matches the upper bound. □

Theorem 5.2 (Failure on the next Hamming shell)**.** *Under the same hypotheses,*

$$\left|\Lambda\left(\mathcal{C}^{\equiv m}, 0, \frac{n-k+1}{n}\right)\right| \geq |\mathbb{F}|^m.$$

Proof. Fix $k-1$ evaluation points $S \subseteq \mathcal{L}$ and set

$$g_S(X) = \prod_{a \in S}(X - a).$$

For every $\lambda = (\lambda_1, \ldots, \lambda_m) \in \mathbb{F}^m$, the tuple

$$(\lambda_1 g_S, \ldots, \lambda_m g_S)$$

defines an interleaved codeword. These $|\mathbb{F}|^m$ codewords are distinct. The zero tuple has distance zero from the received word 0; every nonzero tuple vanishes on S and is nonzero at every evaluation point outside S, so its distance is exactly $n-k+1$. □

Theorem 5.3 (Fractional-target phase split)**.** *Fix $0 < \varepsilon^* < 1$, write $q = |\mathbb{F}|$, and call a radius δ safe when*

$$L_m(\mathcal{C}, \delta) \leq \varepsilon^* q.$$

For every $m \geq 1$, exactly one of the following regimes applies:

(i) *If $\varepsilon^* q < 1$, no radius in $[0,1]$ is safe.*

(ii) *If $1 \le \varepsilon^* q < \binom{n}{k}$, radius zero is safe but the capacity endpoint and every larger radius fail. The optimal sub-endpoint radius is not determined by the present argument.*

(iii) *If $\binom{n}{k} \le \varepsilon^* q$, the largest safe radius on $\mathcal{H}_n$ is*

$$\delta_{\mathrm{cap}} = \frac{n-k}{n} = 1 - \rho;$$

and over real radii the safe set is exactly

$$\left[0, \frac{n-k+1}{n}\right),$$

so it has no largest element and has supremum $(n-k+1)/n$.

Equivalently, the fractional target holds at the capacity endpoint if and only if

$$\binom{n}{k} \le \varepsilon^* |\mathbb{F}|.$$

Proof. Every codeword is contained in the radius-zero ball centered at itself, so no radius can be safe when $\varepsilon^* q < 1$. A radius-zero ball contains at most one codeword, proving its safety when $\varepsilon^* q \ge 1$.

Theorem 5.1 proves both the failure in the intermediate regime and the necessary-and-sufficient endpoint criterion. In the final regime, it and monotonicity of list balls prove safety through count radius $n-k$. A real radius strictly below $(n-k+1)/n$ defines the same ball as some count radius at most $n-k$. At the next grid point, Theorem 5.2 gives a list of size at least $|\mathbb{F}|^m \ge |\mathbb{F}| > \varepsilon^* |\mathbb{F}|$; monotonicity then rules out every larger radius. □

Corollary 5.4 (The explicit smooth family)**.** *For every $u \ge 0$, every $m \ge 1$, and $\mathcal{C}_u = \mathrm{RS}[\mathbb{F}_u, H_u, k_u]$ from Section 4.2, the conclusions of Theorem 5.3*(iii) *hold with $\varepsilon^* = 2^{-128}$. In particular, the largest safe Hamming-grid radius is* $3/4$.

Proof. Apply the inequality proved in Section 4.3. □

5.1 Scope relative to the Grand Challenge

Theorem 5.1 completely determines the endpoint list size for every supplied Reed–Solomon code, smooth or otherwise. Thus the displayed binomial field-size inequality is necessary and sufficient at that endpoint, not merely a convenient sufficient hypothesis. Theorem 5.4 supplies a concrete smooth family in the safe regime. The present argument does not determine the largest sub-endpoint radius for an arbitrary fixed smooth code in the intermediate regime

$$1 \le 2^{-128}|\mathbb{F}| < \binom{n}{k}.$$

The exponential alphabet is what places the explicit family in the final regime; it is not needed for the universal endpoint equality.

This distinction matters in light of known capacity results. Generic Reed–Solomon evaluation points can achieve list-decoding capacity over exponentially large fields with constant list size [BGM24], but generic points do not supply the structured smooth domain required here. Conversely, near-capacity proximity-gap behavior can fail for particular smooth prime-field families [Kam26]. The theorem above imports neither result: it uses only polynomial interpolation and an explicit field-size comparison.

6 The Pair Mutual-Correlated-Agreement Theorem

6.1 Bad challenges

Let $p = (p_0, p_1) \in (\mathbb{F}^2)^I$, and write

$$y_\gamma = p_0 + \gamma p_1.$$

For $S \subseteq I$, say that S is δ-large when

$$\frac{|S|}{n} \ge 1 - \delta.$$

Define $\mathrm{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)$ to be the set of $\gamma \in \mathbb{F}$ for which there exists a δ-large set S such that:

(i) y_γ agrees on S with a codeword of $\mathcal{C}$;

(ii) the pair p does not agree on the same S with a codeword of $\mathcal{C}^{\equiv 2}$.

The MCA error is

$$\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta) = \max_{p \in (\mathbb{F}^2)^I} \frac{|\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)|}{|\mathbb{F}|}.$$

Theorem 6.1 (Uniform pair-MCA certificate bound)**.** *Let $\mathcal{C} = \mathrm{RS}[\mathbb{F}, \mathcal{L}, k]$ have injectively indexed evaluation domain of length n, where $1 \le k \le n$. For every $\delta \in \mathbb{R}$ and every $p \in (\mathbb{F}^2)^n$,*

$$|\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}, \delta, p)| \le \binom{n}{k+1}.$$

Consequently,

$$\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta) \le \frac{\binom{n}{k+1}}{|\mathbb{F}|}.$$

Smoothness is not required for this bound.

Proof. Fix a bad challenge γ, and choose a witnessing set S. Necessarily $|S| > k$. Indeed, when $|S| \le k$, interpolate each component of p on S by a polynomial of degree $< k$; the resulting pair codeword agrees with p on S, contradicting condition (ii).

Choose a k-subset $T \subset S$. Let a_T, b_T be the unique degree-$< k$ polynomials interpolating p_0, p_1, respectively, on T. Condition (ii) says that the pair interpolant (a_T, b_T) fails somewhere on S. Choose such an $i \in S \setminus T$, and set $U = T \cup \{i\}$. Condition (i) remains true after restriction from S to U. The pair still cannot agree with $\mathcal{C}^{\equiv 2}$ on U: any purported pair codeword would agree with (a_T, b_T) on T, hence equal it by k-point determination, contradicting the mismatch at i. Thus U is a $(k+1)$-coordinate obstruction for γ.

A fixed U obstructs at most one challenge. Suppose distinct γ, η were both bad on U, with scalar codewords c_γ, c_η agreeing with $p_0 + \gamma p_1, p_0 + \eta p_1$, respectively. By linearity,

$$d = \frac{c_\gamma - c_\eta}{\gamma - \eta}, \qquad a = c_\gamma - \gamma d$$

are codewords. On U, they agree with p_1 and p_0, respectively, so (a, d) is a pair codeword agreeing with p on U, a contradiction.

Make the finite choices deterministic. Mapping each bad challenge to its unordered obstruction U is injective, and there are $\binom{n}{k+1}$ such subsets. Dividing by $|\mathbb{F}|$ proves the error bound. Notice that the large-set inequality is no longer used after the witness is extracted; this is why the bound is uniform in δ. □

Corollary 6.2 (Explicit smooth family at 2^{-128})**.** *For every member $\mathcal{C}_u$ of the family in Section 4.2, every $\delta \in [0,1]$, and every pair word p,*

$$\frac{|\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}_u, \delta, p)|}{|\mathbb{F}_u|} \le 2^{-128}.$$

Thus, for this family and this same-set pair-MCA definition, every normalized radius is safe and the largest admissible radius is 1*.*

Proof. Section 4.3 proves

$$2^{128}\binom{n_u}{k_u+1} \le q_u.$$

Apply Theorem 6.1. □

6.2 Distance interpretation

Let

$$D_2(p) = \Delta(p, \mathcal{C}^{\equiv 2}), \qquad D_1(y) = \Delta(y, \mathcal{C}).$$

For a challenge outside the MCA bad set, scalar closeness on a large agreement set forces pair closeness on that same set. In distance language, the desired contrapositive has the form

$$D_2(p) \ge \delta \Longrightarrow D_1(y_\gamma) \ge \delta$$

for every MCA-good γ, with the fraction of excluded challenges bounded by $\varepsilon_{\mathrm{MCA}}(\mathcal{C}, \delta)$.

This is a challenge-indexed statement. Replacing the sampled challenge by one fixed field element would require a separate theorem that the selected element is good for every pair, which does not follow from a fractional MCA bound.

6.3 Endpoint interpretation

At $\delta_{\mathrm{cap}} = 1 - k/n$, interpolation on any k coordinates shows that

$$D_1(y) \le \delta_{\mathrm{cap}} \quad \text{and} \quad D_2(p) \le \delta_{\mathrm{cap}}$$

for every scalar word y and pair word p. Moreover, on every set S of size at most k, each component of p extends to a degree-$< k$ polynomial. Such an S can therefore never witness an MCA failure. Every bad endpoint challenge must instead have a witness with $|S| \ge k+1$.

Consequently, the endpoint distance implication above is an equality non-collapse statement: when $D_2(p) = \delta_{\mathrm{cap}}$, a good challenge must not lower the projected distance strictly below δ_{cap}. Theorem 6.1 supplies the previously missing quantitative bound on the excluded challenges.

6.4 Commit-then-challenge order

The mathematical experiment fixes p before sampling $\gamma \leftarrow \mathbb{F}$. This ordering prevents the pair from being selected as a function of the challenge. A full protocol implementation must bind p before deriving γ, but the coding theorem needs only the resulting quantification and probability space; it does not depend on a complete blockchain or commitment protocol.

6.5 Scope

The theorem is universal over pair words and injective RS evaluation domains, but its 2^{-128} corollary uses the explicit exponential-size field. For a different supplied field, the same certificate proof gives the sufficient comparison

$$\binom{n}{k+1} \leq 2^{-128}|\mathbb{F}|.$$

This improves the ordered-certificate requirement by the exact factor $k+1$, since $\binom{n}{k}(n-k) = (k+1)\binom{n}{k+1}$. If that inequality fails, this proof does not determine the optimal radius or show that the MCA target is impossible; it only ceases to certify the requested error.

For an arbitrary supplied finite code, define its exact bad-count spectrum

$$B_{\mathcal{C}}(r) = \max_{p} |\operatorname{Bad}_{\mathrm{MCA}}(\mathcal{C}, r/n, p)|.$$

Then radius r/n meets the 2^{-128} target exactly when

$$2^{128} B_{\mathcal{C}}(r) \leq |\mathbb{F}|.$$

This is also the exact formal equivalence: the Lean theorem identifies the real-valued operational error predicate at normalized radius r/n with the displayed integer inequality, so no rounding relaxation is hidden here. Thus a uniform upper-bound certificate at r, together with one concrete affine line exceeding the budget at $r+1$, pins both the largest safe and first unsafe Hamming-grid radii. The remaining middle regime is the code-specific problem of proving those matching bounds; smoothness by itself does not determine this incidence spectrum.

The result is for the two-coordinate, same-set MCA experiment defined above. Broader notions or higher-dimensional challenge families require separate statements.

7 Application Interface

The coding theorem is stated over the complete received-word space and does not depend on a particular source of observations. An application may nevertheless inherit the result through a distance-aware map.

Let O be a type of raw presented observations, let $G \subseteq O$ be the exact/valid locus, and let

$$\pi : O \longrightarrow (\mathbb{F}^2)^I$$

be the extracted pair word. A source distance $\rho_O : O \to [0,1]$ is meaningful only when O includes arbitrary or noisy presentations; if every element of O already contains a validity witness, then the proximity problem may collapse to the exact code locus.

The Lean companion packages this interface as a source validity model and a source-to-code calibration. Its only quantitative field is the pointwise inequality

$$\rho_O(o) \leq D_2(\pi(o)).$$

It yields, uniformly in δ,

$$\rho_O(o) \geq \delta \Longrightarrow D_2(\pi(o)) \geq \delta.$$

Under this comparison, the proved universal list theorem applies to $\pi(o)$; the proved pair-MCA theorem applies through the same interface once the application fixes its pair before sampling the challenge.

There are two standard ways to discharge the comparison. Defining $\rho_O = D_2 \circ \pi$ makes it reflexive. A genuinely pre-existing source metric instead requires a mathematical proof of $\rho_O \leq D_2 \circ \pi$. Exact witnesses may certify membership in G, but the carrier O must still include arbitrary received observations.

Protocol-specific observation formats, commitments, and runtime verification are intentionally outside the proof of Theorems 5.1, 5.2 and 6.1. Such developments may cite the two coding theorems through the interface above; neither theorem assumes that every received word comes from an application-specific witness space.

8 Formalization And Proof Ledger

8.1 Standalone hierarchy

The intended machine-checked development should separate generic coding mathematics from application bridges:

`Proximity/Core/BlockCode`
: Relative distance, balls, and worst-case list size.

`Proximity/Core/ReedSolomon`
: Smooth-domain RS families and interleaving.

`Proximity/Core/PlanePair`
: Plane–pair–line equivalences.

`Proximity/Core/FiniteThreshold`
: Safe finite grids, greatest-safe and first-unsafe values, and adjacent safe/unsafe threshold certificates.

`Proximity/Capacity/Parameters`
: Explicit ε^* and field-size inequalities.

`Proximity/ListDecoding/Capacity`
: Agreement-set injection, the matching received-word construction, the universal endpoint equality, the fractional-target equivalence, and the next-shell coefficient-vector injection.

`Proximity/MCA/Definitions`
: Arbitrary-block-code same-set MCA definitions at count and relative radii.

`Proximity/MCA/Threshold`
: Exact worst bad-count spectrum and the equivalence between operational real error at r/n and the integer $2^{-\lambda}$ budget.

`Proximity/MCA/SpectrumCertificate`
: Supplied-code grid-threshold pinning from a uniform upper bound at one radius and a concrete over-budget affine line at the next radius.

`Proximity/MCA/Pair`
: RS compressed interpolation certificates, bad-challenge injection, the uniform cardinality bound, and the explicit-family error corollary.

`Proximity/Application/Calibration`

Generic source-validity model, pointwise distance non-collapse, and the uniform source-far to target-far implication.

Existing generic definitions may be imported or refactored, but the final theorem should not depend on HCP, protocol, consensus, or runtime axiom bundles. Both directions of the capacity-grid equality and its exact fractional-target criterion are theorems, not axioms. The next-shell lower bound is also machine-checked. The pair-MCA theorem and its 2^{-128} family instantiation are machine-checked as well.

8.2 Status ledger

Obligation	Status	Required endpoint
Relative Hamming normalization	Lean-defined	Values in $[0, 1]$
Plane–pair equivalence	Lean-proved	Bijection and Hamming isometry
Plane–affine-line equivalence	Lean-proved	Formal inverse maps
Smooth large-field family	Paper-proved	Explicit subgroup family
Capacity-grid list equality	Lean-proved	Universal $L_m = \binom{n}{k}$
Endpoint fractional criterion	Lean-proved	Target iff $\binom{n}{k} \le \varepsilon^*\lvert\mathbb{F}\rvert$
Next-shell list lower bound	Lean-proved	Explicit $\lvert\mathbb{F}\rvert^m$ injection
Pair-MCA certificate bound	Lean-proved	$\lvert \mathrm{Bad}_{\mathrm{MCA}} \rvert \le \binom{n}{k+1}$
MCA target-fraction inequality	Lean-proved	$\binom{n}{k+1} \le \varepsilon^*\lvert\mathbb{F}\rvert$ for the explicit family
Exact supplied-code MCA grid API	Lean-proved	Operational target iff $2^{128}B_{\mathcal{C}}(r) \le \lvert\mathbb{F}\rvert$, with concrete next-line pinning
Explicit-family MCA greatest radius	Lean-proved	Full count radius n, hence normalized radius 1
Largest grid radius / real supremum	Paper-proved	Exact characterization

8.3 Submission boundary

A completed claim requires:

(i) no placeholder assumptions for the main list/MCA result;

(ii) an explicit statement of imported literature theorems;

(iii) all field-size and rounding conditions checked;

(iv) a clear separation between proved theorems, computational evidence, and conjectures;

(v) a reproducible proof artifact and manuscript cross-reference.

8.4 Remaining interfaces

The standalone endpoint list-decoding and pair-MCA packages are closed for the explicit large-field family. The generic source-to-pair calibration

$$d_{\text{source}}(x, \text{valid}) \geq \delta \Longrightarrow \Delta(\pi(x), \mathcal{C}^{\equiv 2}) \geq \delta.$$

is Lean-proved from its named pointwise non-collapse field. Pullback distance makes that field reflexive; an application choosing a different pre-existing source metric supplies one explicit comparison. Separately, the intermediate list-decoding regime still requires new ideas if one wants an optimal sub-endpoint radius for a supplied field that fails the binomial target inequality.

References

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